What's Wrong With Forces

Newton's physics is right. Newton's formulation is a dead end, and this chapter shows you exactly where the road stops.

WHERE WE ARE

Part 0 is finished. Nine chapters, no physics — a derivative that survives curved space, an integral built from accumulation, a vector space, a spectral theorem, a gradient, three field theorems, an oscillator, a Fourier transform. The floor is laid. **This is the first physics chapter of the book**, and from here to the last page nothing is built that isn't used.

Be clear about the stance of what follows, because it is easy to misread. **Newton's physics is correct.** Every prediction $\mathbf{F} = m\mathbf{a}$ makes about a planet, a projectile or a pendulum is right, to the accuracy of the experiment, over the entire domain where the speeds are small and the masses are modest. Nothing in this chapter overturns a single one of those predictions, and Chapter 1.2 will *rederive* all of them.

What fails is the **formulation** — the choice to write the laws in terms of forces and accelerations in Cartesian components. And the failure is not aesthetic. It is structural, and it is fatal in a specific, checkable way: general relativity has no global Cartesian coordinates at all, quantum mechanics has no force operator at all, and field theory has no finite list of particles to sum forces over. Each of the later parts of this book would be *impossible* in the force language. Not harder. Impossible.

So the plan is: state Newton's laws precisely enough that their limits are visible (§1), extract the two things worth keeping — energy (§2) and momentum (§3) — notice a crack in §3 that will not close until Chapter 2.6, then in §4 break the force picture in four separate places, and in §5 look at one small piece of physics that does the same job in a completely different way. That last piece is Fermat's principle, and Chapter 1.2 is nothing but the machinery for taking it seriously.

TOOLS YOU'LL NEED — Chapter 0.1: the derivative as a linear approximation, and $\dot{x}, \ddot{x}$ notation. Chapter 0.2: the Fundamental Theorem, and substitution. Chapter 0.6: partial derivatives, the gradient, and the chain rule along a curve. Chapter 0.7 §2 is the load-bearing one — line integrals, and the four equivalent characterisations of a *conservative* field. This chapter is where that section gets spent. Chapter 0.8: what a second-order ODE is, and the Picard–Lindelöf theorem, which is what turns $\mathbf{F} = m\mathbf{a}$ into a statement about determinism.

1 · Newton's laws, stated precisely

You have seen these. You have probably not seen the fine print, and the fine print is where the interesting failures live.

1.1 · The first law is a definition, not a special case

First law. *There exist frames of reference — called **inertial** — in which a body subject to no net force moves with constant velocity.*

The usual textbook aside is that this is just $\mathbf{F} = m\mathbf{a}$ with $\mathbf{F} = \mathbf{0}$. That reading throws away the entire content. Written as a statement about a particular body, "no force implies no acceleration" is indeed a corollary of the second law and says nothing new. Written as an *existence claim about frames*, it is a separate and much stronger assertion: it says the class of reference frames in which the second law takes its simple form is **non-empty**.

That matters because the second law is false in most frames. Stand in a braking train and every object accelerates backwards with no visible cause. Stand on the rotating Earth and a long pendulum's swing plane creeps around. The second law does not hold in those frames; if you insist on writing it there, you must invent forces — "fictitious" ones — whose only job is to absorb the discrepancy. The first law's function is to say: *there is a class of frames where you don't have to do that, and the second law is a claim about those frames only.*

1.2 · The second law, and what determinism means

Second law. *In an inertial frame, the rate of change of a body's momentum equals the net force on it:*

$$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}, \qquad \mathbf{p} \equiv m\mathbf{v}, \qquad \text{so for constant } m, \qquad \mathbf{F} = m\mathbf{a} = m\ddot{\mathbf{r}}. \tag{1.1.1}$$

Now read (1.1.1) as a piece of mathematics rather than a slogan. It is a **second-order ordinary differential equation** for the function $\mathbf{r}(t)$, of exactly the type Chapter 0.8 was built to handle. Rewrite it as a first-order system in the six-dimensional variable $(\mathbf{r}, \mathbf{v})$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\mathbf{r}\\ \mathbf{v}\end{pmatrix} = \begin{pmatrix}\mathbf{v}\\ \mathbf{F}(\mathbf{r}, \mathbf{v}, t)/m\end{pmatrix}, \tag{1.1.2}$$

and Picard–Lindelöf applies verbatim: if $\mathbf{F}$ is continuous in t and Lipschitz in $(\mathbf{r}, \mathbf{v})$, then specifying $\mathbf{r}$ and $\mathbf{v}$ at one instant determines the entire trajectory, forwards and backwards, uniquely.

> **WHAT DETERMINISM ACTUALLY IS**
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> Classical determinism is not a philosophical posture. It is **the uniqueness half of Picard–Lindelöf, applied to (1.1.1)**. And the reason the state of a classical particle is $(\mathbf{r}, \mathbf{v})$ and not just $\mathbf{r}$ is that the equation is *second* order, so the theorem demands two initial vectors, not one.
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> This is why phase space (Chapter 1.3) has twice the dimension of ordinary space, and it is not a convention: it is forced by the order of the differential equation. Note also that the same theorem will apply, unchanged, to the Schrödinger equation — which is *first* order in time, so a quantum state needs only one initial object. Determinism survives into quantum mechanics; what does not survive is the identification of the state with a position.

1.3 · The third law, in two strengths

Third law (weak form). *If body j exerts a force $\mathbf{F}_{ij}$ on body i, then body i exerts $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$ on body j.*

Third law (strong form). *In addition, $\mathbf{F}_{ij}$ is directed along the line joining the two bodies.*

Gravity and the electrostatic Coulomb force satisfy both. Almost every textbook states only the weak form and then silently uses the strong one, because the weak form gives you momentum conservation and the strong form is what you need for angular momentum. §3 derives both, and then breaks both.

1.4 · What the second law does *not* do

Here is the omission that is almost never said out loud. (1.1.1) does not tell you what $\mathbf{F}$ is. It is an equation with an undetermined symbol in it. Given a force law it predicts motion; without one it predicts nothing at all, and it cannot be tested, because any observed motion whatsoever can be accommodated by *defining* $\mathbf{F} \equiv m\ddot{\mathbf{r}}$.

So Newtonian mechanics is not a theory. It is a **framework awaiting a force law**. The physics lives entirely in the forces, and the forces must be supplied from outside — by Newton's inverse-square gravitation, by Hooke's law, by Coulomb's law, by an empirical drag coefficient. Each of those is a separate empirical input, imported by hand and justified by fitting data.

Hold that thought, because it is the sharpest single argument for what comes next. Chapter 1.2's formulation has the same structure — one function you must supply — but the function is a *scalar*, and by Chapter 1.4 the requirement that it respect a symmetry will constrain it so tightly that in Part VI the force laws stop being inputs and start being *outputs*. That is not available here. You cannot ask "what is the most general force consistent with rotational symmetry?" and get the Standard Model. You can ask it about scalars.

⚠ WHY THIS ISN'T OBVIOUS

The first law's definition of an inertial frame is uncomfortably close to circular. An inertial frame is one where a force-free body moves uniformly; a body is force-free when nothing is pushing it; and how do you know nothing is pushing it? Because it moves uniformly. Newton's own escape was to postulate *absolute space* and define inertial frames as those at rest or in uniform motion relative to it — an entity he admitted was unobservable, and which Mach attacked on exactly that ground.

The circularity is not resolved by being careful. It is resolved by **changing the theory**. In general relativity (Chapter 3.1) the definition is inverted: freely falling bodies *define* the inertial frames, gravity is removed from the list of forces entirely, and "force-free" acquires an independent meaning because a freely falling observer can be identified locally by an experiment (nothing floats away from anything else). The price is that such frames exist only *locally* — over a small enough patch of spacetime — and the failure of local frames to knit together into a global one is precisely the curvature of Chapter 3.4.

So a definition that looks like bookkeeping in Chapter 1.1 turns out to be the seed of the geometric theory of gravity. Worth noticing now that the seam was there.

2 · Energy

Two quantities survive the transition out of the force picture: energy and momentum. Both are extracted from (1.1.1) by a single manipulation, and both will be *redefined* in Chapter 1.4 as consequences of symmetries. Here we get them the ordinary way, because we need them, and because the derivation shows exactly what they cost.

2.1 · The work–energy theorem

Start from $\mathbf{F} = m\dot{\mathbf{v}}$ and dot both sides with $\mathbf{v}$. There is nothing clever about this move; it is the only thing you can do to a vector equation to make a scalar one without choosing a direction by hand.

$$\mathbf{F}\cdot\mathbf{v} \;=\; m\,\dot{\mathbf{v}}\cdot\mathbf{v}. \tag{1.1.3}$$

The right-hand side is a total derivative in disguise. By the product rule applied to a dot product,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{v}\cdot\mathbf{v}) \;=\; \dot{\mathbf{v}}\cdot\mathbf{v}+\mathbf{v}\cdot\dot{\mathbf{v}} \;=\; 2\,\dot{\mathbf{v}}\cdot\mathbf{v}, \qquad \text{so} \qquad m\,\dot{\mathbf{v}}\cdot\mathbf{v} \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\Big(\tfrac{1}{2}mv^2\Big). \tag{1.1.4}$$

The combination $\frac{1}{2}mv^2$ has now appeared on its own, without anyone naming it in advance. Define it:

$$T \;\equiv\; \tfrac{1}{2}mv^2 \qquad (\textbf{kinetic energy}). \tag{1.1.5}$$

Then (1.1.3) reads $\mathbf{F}\cdot\mathbf{v} = \mathrm{d}T/\mathrm{d}t$. Integrate both sides in time from t_1 to t_2 along the actual trajectory $\mathbf{r}(t)$, and recognise the left-hand side using Chapter 0.7's parametrised line integral, since $\mathbf{v}\,\mathrm{d}t = \mathrm{d}\mathbf{r}$:

$$\underbrace{\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}}_{W} \;=\; \int_{t_1}^{t_2}\mathbf{F}\cdot\mathbf{v}\,\mathrm{d}t \;=\; T(t_2)-T(t_1) \;=\; \Delta T. \tag{1.1.6}$$

This is the **work–energy theorem**, and note how little it assumed: no property of $\mathbf{F}$ whatsoever. It holds for friction, for magnetic forces, for a force you invent on the spot. The work done *along the actual path* equals the change in kinetic energy. Always.

It also does not yet give you a conservation law, because the left-hand side is a line integral along a path and there is no reason for it to be expressible as a difference of anything. Getting a conservation law requires an extra hypothesis, and Chapter 0.7 already told us precisely which one.

2.2 · Conservative forces — cashing Chapter 0.7

Chapter 0.7 §2.3 listed four conditions on a vector field $\mathbf{F}$ defined on a region U and proved that on a **simply connected** domain all four are the same condition:

	CONDITION	CHARACTER
(A)	$\mathbf{F} = -\nabla V$ for a single-valued function V on U	a potential exists
(B)	$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ depends only on the endpoints of C	path-independence
(C)	$\oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = 0$ for every closed loop	no free energy from a cycle
(D)	$\nabla \times \mathbf{F} = \mathbf{0}$ throughout U	a *local* test

(The minus sign in (A) is a convention chosen so that V decreases in the direction the force pushes; Chapter 0.7 wrote $\mathbf{F} = \nabla\phi$, so $V = -\phi$.) A force satisfying these is **conservative**, and the name is about to be earned rather than assumed.

The equivalence is the whole point, so do not let it slide past. (D) is three derivatives evaluated at a point and takes ten seconds. (B) and (C) are statements about infinitely many paths and cannot be checked directly. (A) is what you actually want, because it replaces a vector field by a single number at each point. That you may move freely between a ten-second local test and a global existence statement is the entire practical value of Chapter 0.7, and it is worth 2π of caution: the arrow (D)⇒(C) fails on a domain that is not simply connected, which is what the punctured-plane counterexample there was for.

Now suppose $\mathbf{F}$ is conservative. Then by (A) and Chapter 0.7's gradient theorem,

$$W \;=\; \int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} \;=\; -\int_C \nabla V \cdot \mathrm{d}\mathbf{r} \;=\; -\big[V(\mathbf{r}_2) - V(\mathbf{r}_1)\big] \;=\; -\Delta V. \tag{1.1.7}$$

Set this equal to ΔT from (1.1.6):

$$\begin{aligned} \Delta T \;=\; -\Delta V \quad &\Longleftrightarrow \quad \Delta\big(T + V\big) \;=\; 0 \\ &\Longleftrightarrow \quad \boxed{E \;\equiv\; T + V \;=\; \text{constant.}} \end{aligned} \tag{1.1.8}$$

Energy conservation, derived — and derived in a way that shows exactly what it rests on. The quantity V is called the **potential energy**, and it exists because of (A); the conservation holds because of (B); the reason you can verify it in practice is (D). Three faces of one condition, all three used, all three necessary.

FAMILIAR GROUND — PATH FUNCTIONS AND STATE FUNCTIONS

The distinction in §2.2 is one you already reason with daily, in a domain with no vectors in it.

Cumulative anthracycline exposure is a path function. It is a line integral of dose rate along the entire treatment history, $D = \int_\gamma \dot{d}\,\mathrm{d}t$, and it is why the quantity that predicts cardiotoxicity is *lifetime cumulative dose* — not the current dose, not the current plasma concentration, not the dose per cycle. Two patients standing in front of you in identical present states, same echocardiogram, same current regimen, can carry different values of D because they got here along different routes. You cannot read D off the present state; you have to know the history.

A **state function** is the opposite: a quantity ϕ attached to the present state alone, so that its change over any course of events is $\phi(\text{end}) - \phi(\text{start})$ and the route is irrelevant. Serum creatinine is like this; total fluid administered is not.

That is exactly conditions (B) and (A). A conservative field has a potential, and its line integral collapses to an endpoint difference:

$$\int_\gamma \mathbf{F}\cdot\mathrm{d}\mathbf{r} \;=\; -\big[V(\text{end}) - V(\text{start})\big] \qquad \text{— route-independent.}$$

A non-conservative field has no potential and its line integral does not collapse; you must carry the whole path. The mathematical content is only this: **when is $\int_\gamma$ replaceable by $\phi(\text{end}) - \phi(\text{start})$?** The answer, identically in both settings, is *when the integrand is the derivative of something.* This is also why thermodynamics insists on writing $\mathrm{d}U$ but δQ and δW: internal energy is a state function, heat and work are path functions, and the notation is carrying condition (A).

▼ **Grind box — where friction's energy went, and a first look at coarse-graining**

Take a projectile with quadratic drag: $\mathbf{F} = -mg\hat{\mathbf{y}} - k\,|\mathbf{v}|\,\mathbf{v}$. The drag term is not conservative — its work depends on the path, and a closed loop returns negative work, violating (C). Integrate the motion numerically ($m = 1.7$, $k = 0.42$, $g = 9.81$, launched from the origin at $\mathbf{v}_0 = (9, 14)$, run to $t = 2.4$) and check the two statements separately:

QUANTITY	VALUE
$\int \mathbf{F}\cdot\mathbf{v}\,\mathrm{d}t$ (all forces)	-202.4463642316
ΔT	-202.4463642317
$\Delta(T + mgy)$	-282.6869372
work done by drag alone	-282.6869372

The work–energy theorem (1.1.6) holds to eleven digits, as it must — it assumed nothing. But $T + V$ is *not* conserved, and the deficit is exactly the work done by the non-conservative force. (1.1.8) failed, and it failed at precisely the step that used (A).

Now the physical question: is energy conservation actually violated? No. The missing 282.69 J is in the air and the projectile as thermal energy — the kinetic energy of $\sim 10^{25}$ molecules whose individual positions and velocities you declined to track. Every one of those microscopic degrees of freedom obeys conservative electromagnetic forces. Friction is not a violation of energy conservation; it is a **bookkeeping choice**. You threw away 10^{25} coordinates and the energy went with them.

That is worth stating as a principle, because it recurs at every scale of this book: **non-conservative forces are what conservative forces look like after you stop tracking some of the degrees of freedom.** The technical name for declining to track them is *coarse-graining*. And the asymmetry it introduces — you can predict where the energy goes but not get it back — is the origin of the second law of thermodynamics, which is a statement about the coarse-graining and not about the microscopic dynamics, all of which is time-reversible. The same move reappears in Chapter 5.9 under the name *renormalization group*: integrate out the degrees of freedom you cannot resolve, and the effective description of what is left acquires properties the microscopic one did not have.

⚠ WHY THIS ISN'T OBVIOUS

"Energy is conserved" is presented in school as a law of nature that holds always. As derived here it is nothing of the kind: it is a *theorem with a hypothesis*, and the hypothesis is that every force in play is conservative. Drop the hypothesis and (1.1.8) simply does not follow — as the grind box measures.

Two corrections to the school version, in opposite directions. **Downward:** for the system you chose to write down, $T + V$ genuinely is not conserved when friction acts, and pretending otherwise gives wrong answers. **Upward:** the reason it looks like a universal law anyway is that the microscopic forces really are conservative, so a large enough accounting always closes.

And the deepest version of the statement is not about forces at all. Chapter 1.4 proves **Noether's theorem**: energy is conserved if and only if the laws of physics do not change with time. That is a much better answer to "why is energy conserved", and it has a startling consequence — in an expanding universe the laws *are* time-dependent, and the energy of the cosmological photon gas is correspondingly *not* conserved (Chapter 3.8). A quantity whose conservation you have been taught is absolute turns out to be contingent on a symmetry that the universe does not exactly possess. You cannot get to that sentence from $\mathbf{F} = m\mathbf{a}$.

3 · Momentum, angular momentum — and a crack

3.1 · Total momentum, from the weak third law

Take N particles. Particle i feels an external force $\mathbf{F}_i^{\text{ext}}$ plus internal forces $\mathbf{F}_{ij}$ from each of the others, so (1.1.1) reads $\dot{\mathbf{p}}_i = \mathbf{F}_i^{\text{ext}} + \sum_{j\neq i} \mathbf{F}_{ij}$. Define the total momentum $\mathbf{P} = \sum_i \mathbf{p}_i$ and add:

$$\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} = \sum_i \mathbf{F}_i^{\text{ext}} + \sum_i \sum_{j\neq i} \mathbf{F}_{ij}. \tag{1.1.9}$$

The double sum runs over ordered pairs, so each unordered pair $\{i, j\}$ appears twice, as $\mathbf{F}_{ij}$ and $\mathbf{F}_{ji}$. By the weak third law those two cancel. Every pair cancels; the double sum is zero; and

$$\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} = \mathbf{F}_{\text{total}}^{\text{ext}}, \qquad \text{so} \qquad \mathbf{F}_{\text{total}}^{\text{ext}} = \mathbf{0} \implies \mathbf{P} = \text{constant}. \tag{1.1.10}$$

Internal forces, however violent, cannot move the total momentum. This is why a rocket works and why the centre of mass of an exploding shell continues on its parabola.

3.2 · Angular momentum, from the strong third law

Define $\mathbf{L} = \sum_i \mathbf{r}_i \times \mathbf{p}_i$ and differentiate. The term $\dot{\mathbf{r}}_i \times \mathbf{p}_i = \mathbf{v}_i \times m_i \mathbf{v}_i$ vanishes because a vector's cross product with itself is zero, leaving

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \sum_i \mathbf{r}_i \times \dot{\mathbf{p}}_i = \underbrace{\sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{ext}}}_{\boldsymbol{\tau}^{\text{ext}}} + \sum_i \sum_{j\neq i} \mathbf{r}_i \times \mathbf{F}_{ij}. \tag{1.1.11}$$

Pair up the internal terms again. Using $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$, the pair $\{i, j\}$ contributes

$$\mathbf{r}_i \times \mathbf{F}_{ij} + \mathbf{r}_j \times \mathbf{F}_{ji} = (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{F}_{ij}, \tag{1.1.12}$$

which vanishes *if and only if* $\mathbf{F}_{ij}$ is parallel to $\mathbf{r}_i - \mathbf{r}_j$ — the line joining the two particles. That is exactly the strong form. So with it, $\mathrm{d}\mathbf{L}/\mathrm{d}t = \boldsymbol{\tau}^{\text{ext}}$, and an isolated system conserves angular momentum. Without it, the internal forces can spin the system up out of nothing.

▼ **Grind box — why the weak form is not enough, with a concrete counterexample**

Take two particles at $\mathbf{r}_1 = (0, 0, 0)$ and $\mathbf{r}_2 = (d, 0, 0)$, and suppose they exert on each other forces that are equal and opposite but *not* along the line joining them: $\mathbf{F}_{12} = f\hat{\mathbf{y}}$ and $\mathbf{F}_{21} = -f\hat{\mathbf{y}}$. The weak third law holds exactly, so by (1.1.10) the total momentum is conserved.

Now compute the internal torque from (1.1.12):

$$\big(\mathbf{r}_1 - \mathbf{r}_2\big) \times \mathbf{F}_{12} \;=\; (-d\hat{\mathbf{x}}) \times (f\hat{\mathbf{y}}) \;=\; -df\,\hat{\mathbf{z}} \;\neq\; \mathbf{0}.$$

The pair develops angular momentum with no external agent. If you could build such a force you would have a reactionless flywheel: the system spins faster and faster, and since rotational kinetic energy grows as L^2, you could extract unbounded work. The strong third law is what forbids this, and it is a genuinely separate assumption — it does not follow from the weak one, as this example shows in three lines.

Note also which symmetry each form is standing in for. The weak form is doing the job that Chapter 1.4 will assign to *translational* invariance, and the strong form the job of *rotational* invariance. In the Lagrangian formulation you assume the symmetries and the conservation laws follow; here you assume the conservation-law-shaped force conditions and hope they hold. The next section shows that one of them does not.

3.3 · The crack: the third law is false

Everything in §3.1 and §3.2 rests on the third law, which is an assumption about forces, not a theorem. It is worth asking whether any real force violates it. One does, and it is not exotic — it is the magnetic force between two moving charges.

We need two ingredients, both of which this book will derive in Chapter 2.6 and both of which are quoted here.

⚑ QUOTED, NOT PROVED — THE LORENTZ FORCE AND THE FIELD OF A MOVING CHARGE

A charge q moving with velocity $\mathbf{v}$ through fields $\mathbf{E}, \mathbf{B}$ feels $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. A charge q moving slowly with velocity $\mathbf{v}$ produces, at displacement $\mathbf{r}$ from itself, the magnetic field

$$\mathbf{B}(\mathbf{r}) \;=\; \frac{\mu_0}{4\pi}\,\frac{q\,\mathbf{v} \times \hat{\mathbf{r}}}{|\mathbf{r}|^2}, \qquad \hat{\mathbf{r}} = \mathbf{r}/\,|\mathbf{r}|\,,$$

which is the Biot–Savart law for a point charge, valid to first order in v/c. Both are derived in Chapter 2.6 as consequences of relativity applied to Coulomb's law; neither is assumed anywhere else in this chapter.

Now the configuration, chosen to make the failure as stark as possible. Put charge q_1 at the origin moving with $\mathbf{v}_1 = v\hat{\mathbf{x}}$, and charge q_2 at $\mathbf{r}_2 = d\hat{\mathbf{x}}$ — directly ahead of it — moving *perpendicular* to that, with $\mathbf{v}_2 = v\hat{\mathbf{y}}$.

Force on 2 from 1. The displacement from 1 to 2 is $d\hat{\mathbf{x}}$, so $\hat{\mathbf{r}} = \hat{\mathbf{x}}$, and

$$\mathbf{B}_1(\mathbf{r}_2) \;=\; \frac{\mu_0 q_1}{4\pi d^2}\,\big(v\hat{\mathbf{x}}\big) \times \hat{\mathbf{x}} \;=\; \mathbf{0}, \tag{1.1.13}$$

because charge 1 is moving straight at charge 2 and a charge produces no magnetic field directly ahead of itself. So $\mathbf{F}_{21}^{\mathrm{mag}} = q_2\mathbf{v}_2 \times \mathbf{B}_1 = \mathbf{0}$. *Charge 2 feels no magnetic force at all.*

Force on 1 from 2. The displacement from 2 to 1 is $-d\hat{\mathbf{x}}$, so now $\hat{\mathbf{r}} = -\hat{\mathbf{x}}$ and

$$\mathbf{B}_2(\mathbf{r}_1) = \frac{\mu_0 q_2}{4\pi d^2}\,(v\hat{\mathbf{y}}) \times (-\hat{\mathbf{x}}) = \frac{\mu_0 q_2 v}{4\pi d^2}\,\hat{\mathbf{z}}, \tag{1.1.14}$$

using $\hat{\mathbf{y}} \times \hat{\mathbf{x}} = -\hat{\mathbf{z}}$. This is not zero, and the magnetic force on charge 1 is

$$\mathbf{F}_{12}^{\mathrm{mag}} = q_1\mathbf{v}_1 \times \mathbf{B}_2 = \frac{\mu_0 q_1 q_2 v^2}{4\pi d^2}\,(\hat{\mathbf{x}} \times \hat{\mathbf{z}}) = -\frac{\mu_0 q_1 q_2 v^2}{4\pi d^2}\,\hat{\mathbf{y}}. \tag{1.1.15}$$

The electric forces between the two charges are equal and opposite along the line joining them, so they cancel in the sum. What is left is the magnetic residue:

$$\mathbf{F}_{12} + \mathbf{F}_{21} = -\frac{\mu_0 q_1 q_2 v^2}{4\pi d^2}\,\hat{\mathbf{y}} \neq \mathbf{0}. \tag{1.1.16}$$

The third law fails. Not approximately, not in some limit — one particle is pushed and the other is not. And the strong form fails twice over, since the leftover force is *perpendicular* to the line joining the charges, so (1.1.12) delivers a net internal torque as well.

> ⚑ **ONE TERM WE DROPPED, AND WHY IT DOES NOT MATTER HERE**
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> The step above — "the electric forces cancel in the sum" — used the *static* Coulomb field. The electric field of a charge in motion is not quite Coulomb's: it is compressed transversely, and the correction enters at order v^2/c^2, exactly the order of the magnetic residue we just computed. Keeping it leaves an additional uncancelled piece along $\hat{\mathbf{x}}$, so the true failure is somewhat larger than (1.1.16) and points in a different direction.
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> That does not weaken the argument — it strengthens it, and we drop the term only because the point being made is qualitative: *some* momentum goes missing. Chapter 2.6 derives the moving charge's field properly, computes both components, and then shows where the missing momentum actually is. The bookkeeping there balances identically.

Now feed that into (1.1.10). There is nothing outside this system — two charges alone in the universe — and yet

$$\frac{\mathrm{d}\mathbf{P}_{\mathrm{mech}}}{\mathrm{d}t} = -\frac{\mu_0 q_1 q_2 v^2}{4\pi d^2}\,\hat{\mathbf{y}} \neq \mathbf{0}. \tag{1.1.17}$$

The total mechanical momentum of an isolated system changes. Momentum is appearing from nowhere, at a computable rate. Let us be honest about how big the effect is before deciding how seriously to take it: comparing (1.1.16) to the Coulomb force $q_1 q_2/4\pi\varepsilon_0 d^2$ and using $\mu_0\varepsilon_0 = 1/c^2$,

$$\frac{|\mathbf{F}^{\text{net}}|}{F_{\text{Coulomb}}} = \mu_0\varepsilon_0 v^2 = \frac{v^2}{c^2}, \tag{1.1.18}$$

which for $v = 10^6\ \text{m s}^{-1}$ is 1.11×10^{-5} — small, but not zero, and not subtle at all in a particle accelerator. (Both the ratio and the vector directions above were checked numerically; the computed ratio is 1.112650×10^{-5} against $v^2/c^2 = 1.112650 \times 10^{-5}$.)

EITHER MOMENTUM CONSERVATION IS FALSE, OR MOMENTUM LIVES SOMEWHERE ELSE

There are exactly two ways out of (1.1.17), and they are not equally attractive.

One: momentum conservation is simply false, and the deep symmetry between "total momentum is constant" and "space looks the same everywhere" is a coincidence that breaks down whenever charges move. Nobody has ever found an experiment supporting this, and it would wreck the structure of Chapter 1.4.

Two: momentum is conserved, and the missing amount is stored in something that is not a particle. Since the only other object in the problem is the electromagnetic field, **the field carries momentum.** Chapter 2.6 will derive its density, $\mathbf{g} = \varepsilon_0 \mathbf{E} \times \mathbf{B}$, show that $\mathbf{P}_{\text{mech}} + \mathbf{P}_{\text{field}}$ is exactly conserved, and identify the corresponding energy flux as the Poynting vector.

The second is right, and its consequence is worth pausing on. Up to this point in your physics education a "field" has plausibly been a bookkeeping device — a map of where the force would be if you put a test charge there, with no independent existence. (1.1.17) kills that reading. Something that stores momentum, that you can push on and that pushes back, that can carry momentum across a room in the gap between one charge slowing down and another speeding up, is not a bookkeeping device. **It is an object.** This is the first place in the book where a field is forced to be real, and it is why Part V ends up treating fields as more fundamental than the particles that were supposed to be sourcing them.

Notice also what has just happened to the force picture, quietly. Newtonian mechanics is a theory of finitely many particles exerting instantaneous forces on one another at a distance. Special relativity forbids instantaneous anything. The moment you take the delay seriously — charge 1 moved, and charge 2 will not learn about it for d/c seconds — you must ask where the momentum *is* during the interval, and the only possible answer is "in the field". The third law's failure is relativity's opening move, arriving a full part of the book early.

4 · Where the force picture actually breaks

Four failures. None of them is a matter of taste, and each is demonstrated rather than asserted.

4.1 · Failure one: constraints

A plane pendulum: a bob of mass m on a string of fixed length ℓ from a pivot, swinging in a vertical plane. This is the simplest constrained system in physics. Let us solve it with Newton, in Cartesian coordinates, without cheating.

Put the origin at the pivot, $\hat{\mathbf{x}}$ horizontal, $\hat{\mathbf{y}}$ vertically upward. The bob is at (x, y) and there are two forces on it: gravity $(0, -mg)$, and the string tension, which is directed from the bob *toward the pivot*, i.e. along $-(x, y)/\ell$. Call its magnitude $\mathcal{F}$ — the letter T is spoken for by (1.1.5) — and note the critical fact: $\mathcal{F}$ **is unknown.** It is not given by any force law. A string pulls with whatever tension is required to keep its length fixed, and how much that is depends on the motion, which is what we are trying to find.

Newton's second law in components:

$$m\ddot{x} \;=\; -\frac{\mathcal{F}}{\ell}\,x, \qquad\qquad m\ddot{y} \;=\; -\frac{\mathcal{F}}{\ell}\,y \;-\; mg. \tag{1.1.19}$$

Two equations, three unknown functions $x(t), y(t), \mathcal{F}(t)$. The system is underdetermined and Picard–Lindelöf does not apply. The missing information is the constraint

$$x^2 + y^2 \;=\; \ell^2, \tag{1.1.20}$$

which is an *algebraic* relation, not a differential equation, so it cannot simply be appended to (1.1.19). To use it we must differentiate it into a compatible form. Once:

$$2x\dot{x} + 2y\dot{y} = 0 \qquad\Longleftrightarrow\qquad x\dot{x} + y\dot{y} = 0, \tag{1.1.21}$$

which says the velocity is perpendicular to the string — true, and not yet enough, because it involves no accelerations. Differentiate again, by the product rule:

$$\dot{x}^2 + x\ddot{x} + \dot{y}^2 + y\ddot{y} = 0 \qquad\Longleftrightarrow\qquad x\ddot{x} + y\ddot{y} \;=\; -\big(\dot{x}^2 + \dot{y}^2\big). \tag{1.1.22}$$

Now we have a relation among accelerations and can combine it with (1.1.19). Multiply the first equation of (1.1.19) by x, the second by y, and add:

$$m\big(x\ddot{x} + y\ddot{y}\big) \;=\; -\frac{\mathcal{F}}{\ell}\big(x^2 + y^2\big) - mgy \;=\; -\mathcal{F}\ell - mgy, \tag{1.1.23}$$

using (1.1.20) in the last step. Substituting (1.1.22) on the left and solving for the tension:

$$\boxed{\mathcal{F} \;=\; \frac{m}{\ell}\Big(\dot{x}^2 + \dot{y}^2 \;-\; g\,y\Big).} \tag{1.1.24}$$

We have determined the unknown force — but only in terms of the motion, which we still do not know. Put (1.1.24) back into (1.1.19) to close the system:

$$\begin{aligned} \ddot{x} &= -\frac{x}{\ell^2}\Big(\dot{x}^2 + \dot{y}^2 - gy\Big), \\ \ddot{y} &= -\frac{y}{\ell^2}\Big(\dot{x}^2 + \dot{y}^2 - gy\Big) \ - \ g. \end{aligned} \tag{1.1.25}$$

Look at what we have. A pair of coupled nonlinear second-order ODEs in two variables, for a system that visibly has *one* degree of freedom. Nothing about (1.1.25) suggests a pendulum. You cannot see the period in it, you cannot see the small-oscillation limit, and you cannot integrate it in this form. It is also numerically treacherous: the constraint (1.1.20) is only enforced through its second derivative, so any integration error accumulates and the bob slowly drifts off its circle.

So do the thing we should have done at the start and introduce the angle θ from the downward vertical:

$$x = \ell\sin\theta, \qquad y = -\ell\cos\theta, \tag{1.1.26}$$

which satisfies (1.1.20) automatically. Differentiate twice:

$$\begin{aligned} \dot{x} &= \ell\cos\theta\,\dot{\theta}, \quad & \ddot{x} &= \ell\cos\theta\,\ddot{\theta} - \ell\sin\theta\,\dot{\theta}^2, \\ \dot{y} &= \ell\sin\theta\,\dot{\theta}, \quad & \ddot{y} &= \ell\sin\theta\,\ddot{\theta} + \ell\cos\theta\,\dot{\theta}^2. \end{aligned} \tag{1.1.27}$$

Then $\dot{x}^2 + \dot{y}^2 = \ell^2\dot{\theta}^2(\cos^2\theta + \sin^2\theta) = \ell^2\dot{\theta}^2$, and $-gy = g\ell\cos\theta$, so the bracket in (1.1.25) is $\ell^2\dot{\theta}^2 + g\ell\cos\theta$. Substituting into the first line of (1.1.25):

$$\begin{aligned} \ell\cos\theta\,\ddot{\theta} - \ell\sin\theta\,\dot{\theta}^2 \ &= \ -\frac{\ell\sin\theta}{\ell^2}\Big(\ell^2\dot{\theta}^2 + g\ell\cos\theta\Big) \\ &= \ -\ell\sin\theta\,\dot{\theta}^2 - g\sin\theta\cos\theta. \end{aligned} \tag{1.1.28}$$

The $\ell\sin\theta\,\dot{\theta}^2$ terms cancel on both sides, leaving $\ell\cos\theta\,\ddot{\theta} = -g\sin\theta\cos\theta$, and dividing by $\ell\cos\theta$ (legitimate whenever $\cos\theta \neq 0$):

$$\boxed{\ddot{\theta} \ = \ -\frac{g}{\ell}\sin\theta.} \tag{1.1.29}$$

And the tension, from (1.1.24), is $\mathcal{F} = m\big(\ell\dot{\theta}^2 + g\cos\theta\big)$ — the centripetal requirement plus the component of weight along the string.

An experienced hand would not have gone this way. Instead of Cartesian components you resolve the two forces *along and perpendicular to the string*, which are the directions the geometry actually prefers. Perpendicular to the string (tangentially), the tension contributes nothing at all, gravity contributes $-mg\sin\theta$, and the tangential acceleration is

$\ell\ddot{\theta}$; along the string (radially inward) the acceleration is the centripetal $\ell\dot{\theta}^2$, and the inward forces are $\mathcal{F}$ less the inward component $mg\cos\theta$ of the weight:

$$m\ell\ddot{\theta} = -mg\sin\theta, \qquad\qquad m\ell\dot{\theta}^2 = \mathcal{F} - mg\cos\theta. \tag{1.1.30}$$

Two component equations, two unknowns $(\theta, \mathcal{F})$, and the tension is eliminated by the simple expedient of ignoring the second — which exists only to determine a quantity we did not want and will now discard. That is quicker, and it is what anyone competent actually does. But look at what it presupposed: you had to know *in advance* that the acceleration in polar coordinates splits into a tangential $\ell\ddot{\theta}$ and a centripetal $\ell\dot{\theta}^2$. That is not a guess; it is (1.1.35), which §4.2 spends half a page deriving. The Cartesian route above is longer precisely because it assumes nothing. Both routes carry an unknown force through to the end and then throw it away. (Both results were checked by integrating (1.1.25) and (1.1.29) numerically from the same initial condition: the trajectories agree to 3×10^{-11} over seven seconds, and a bob released from horizontal has $\mathcal{F} = 3.000\,mg$ at the bottom of its swing.)

▼ Grind box — the second component equation, and what its redundancy is telling you

We used only the x-equation of (1.1.25). Honesty requires checking the y-equation too. Substituting (1.1.27):

$$\begin{aligned} \ell\sin\theta\,\ddot{\theta} + \ell\cos\theta\,\dot{\theta}^2 &= \frac{\ell\cos\theta}{\ell^2}\Big(\ell^2\dot{\theta}^2 + g\ell\cos\theta\Big) - g \\ &= \ell\cos\theta\,\dot{\theta}^2 + g\cos^2\theta - g \;=\; \ell\cos\theta\,\dot{\theta}^2 - g\sin^2\theta, \end{aligned}$$

using $\cos^2\theta - 1 = -\sin^2\theta$. The $\ell\cos\theta\,\dot{\theta}^2$ terms cancel and we are left with $\ell\sin\theta\,\ddot{\theta} = -g\sin^2\theta$, i.e. $\ddot{\theta} = -(g/\ell)\sin\theta$ again, valid whenever $\sin\theta \neq 0$.

The two equations are not independent. Each degenerates to $0 = 0$ somewhere — the x-equation at $\theta = \pm\pi/2$, the y-equation at $\theta = 0, \pi$ — and between them they always deliver the same single equation of motion. That redundancy is the constraint speaking: two Cartesian equations minus one constraint equals one genuine equation, and the system was one-dimensional from the start. We paid for two coordinates, carried an unknown force through five steps, and then had to *discover* that we had been solving one equation all along.

There is a slicker elimination, worth seeing because it shows how much depends on spotting the right trick. Multiply the first equation of (1.1.19) by y, the second by x, and subtract; the tension terms are $-\mathcal{F}xy/\ell$ in both and cancel without ever being evaluated:

$$m\big(y\ddot{x} - x\ddot{y}\big) \;=\; mgx \qquad\Longrightarrow\qquad y\ddot{x} - x\ddot{y} = gx.$$

Substituting (1.1.27), the left side collapses to $-\ell^2\ddot{\theta}$ (the $\dot{\theta}^2$ terms cancel identically) and the right to $g\ell\sin\theta$, giving (1.1.29) in one line. Elegant — and entirely dependent on noticing that $y\times(\text{first}) - x\times(\text{second})$ kills the unknown. That combination is, not coincidentally, the z-component of the torque about the pivot; you have rediscovered angular momentum, by hand, because Cartesian components hid it. Chapter 1.4 makes that systematic.

One more piece of fine print. Differentiating (1.1.20) twice enlarges the solution set: (1.1.22) is implied by the constraint but does not imply it. The general solution of the differentiated system has $x^2 + y^2 = \ell^2 + at + b$ for constants fixed by initial data, so you must impose (1.1.20) *and* (1.1.21) at $t = 0$ to kill a and b. Forget either and you are integrating the wrong problem. This is the standard hazard of index reduction in constrained dynamics, it is a real source of bugs in physics engines, and Chapter 1.2 does not encounter it, because there is no constraint left to differentiate.

COUNT THE COST

To get (1.1.29) — one line of physics — we needed: two component equations; one unknown force that appears in neither the question nor the answer; a constraint differentiated twice, with spurious solutions to exclude; an elimination; a coordinate change; and a redundancy to check. Roughly a page.

Chapter 1.2 §6.2 obtains the identical equation in three lines, from a single scalar function, in which the tension never appears at all — because a force perpendicular to the motion does no work, and the new formulation is built out of work. Turn to it after §5 and count the lines yourself. The comparison is the reason Part I exists.

4.2 · Failure two: coordinates, and the loss of form invariance

The pendulum hinted at it; this makes it precise. Take $\mathbf{F} = m\mathbf{a}$ in the plane and write it in polar coordinates (r, θ), which is the natural choice for any central force. The result is famous and disturbing.

Everything hinges on the fact that the polar basis vectors *point in different directions at different places*:

$$\hat{\mathbf{r}} = \big(\cos\theta,\ \sin\theta\big), \qquad \hat{\boldsymbol{\theta}} = \big(-\sin\theta,\ \cos\theta\big). \tag{1.1.31}$$

Along a trajectory θ changes with time, so these vectors are themselves functions of time and must be differentiated. By the chain rule,

$$\dot{\hat{\mathbf{r}}} = \dot\theta\,\big(-\sin\theta,\ \cos\theta\big) = \dot\theta\,\hat{\boldsymbol{\theta}}, \qquad \dot{\hat{\boldsymbol{\theta}}} = \dot\theta\,\big(-\cos\theta,\ -\sin\theta\big) = -\dot\theta\,\hat{\mathbf{r}}. \tag{1.1.32}$$

Now differentiate the position $\mathbf{r} = r\,\hat{\mathbf{r}}$ twice, using the product rule and (1.1.32) at each step:

$$\mathbf{v} \;=\; \dot r\,\hat{\mathbf{r}} + r\,\dot{\hat{\mathbf{r}}} \;=\; \dot r\,\hat{\mathbf{r}} + r\dot\theta\,\hat{\boldsymbol{\theta}}, \tag{1.1.33}$$

$$\begin{aligned} \mathbf{a} \;=\; \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} \;&=\; \ddot r\,\hat{\mathbf{r}} + \dot r\,\dot{\hat{\mathbf{r}}} + \big(\dot r\dot\theta + r\ddot\theta\big)\hat{\boldsymbol{\theta}} + r\dot\theta\,\dot{\hat{\boldsymbol{\theta}}} \\ &=\; \ddot r\,\hat{\mathbf{r}} + \dot r\dot\theta\,\hat{\boldsymbol{\theta}} + \big(\dot r\dot\theta + r\ddot\theta\big)\hat{\boldsymbol{\theta}} - r\dot\theta^2\,\hat{\mathbf{r}}. \end{aligned} \tag{1.1.34}$$

Collect components. Newton's law $\mathbf{F} = m\mathbf{a}$ in polar coordinates therefore reads

$$\boxed{m\left(\ddot{r} - r\dot{\theta}^2\right) = F_r, \qquad m\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) = F_\theta.} \tag{1.1.35}$$

Stare at this. In Cartesian coordinates the law was $m\ddot{x} = F_x, m\ddot{y} = F_y$ — mass times second derivative equals force, one term each. In polar coordinates two extra terms have materialised: $-mr\dot{\theta}^2$, the **centrifugal** term, and $2m\dot{r}\dot{\theta}$, the **Coriolis** term.

Where did they come from? Not from any force: we changed no physics, added no interaction, and $\mathbf{F}$ is the same vector it always was. They came from (1.1.32) — from the basis vectors turning. They are **artefacts of the coordinate system**.

Here is the cleanest demonstration that they are artefacts. Take a completely free particle, $\mathbf{F} = \mathbf{0}$, moving in a straight line: $x = v_0 t, y = b$ with $b \neq 0$. Then $r = \sqrt{b^2 + v_0^2 t^2}$, and differentiating twice gives

$$\ddot{r} = \frac{b^2 v_0^2}{\left(b^2 + v_0^2 t^2\right)^{3/2}} \neq 0. \tag{1.1.36}$$

A particle with no force on it has a nonzero second derivative of its radial coordinate — at $v_0 = b = 1,\ t = 1$ the value is 0.35355. It is exactly cancelled by $r\dot{\theta}^2$, which at the same instant is also 0.35355, so that (1.1.35) gives $F_r = 0$ as it must. But "$m\ddot{r} = F_r$" would have been flatly wrong.

$\mathbf{F} = M\mathbf{A}$ IS NOT FORM INVARIANT

A law is **form invariant** under a class of coordinate changes if it has the same *shape* in all of them. $\mathbf{F} = m\mathbf{a}$ is not. Its shape — literally, which terms appear — is correct in Cartesian coordinates and wrong in every other coordinate system, where it must be patched with terms whose values depend on the labelling and not on the physics.

You can live with this in the plane. There are only a handful of useful coordinate systems and their correction terms are tabulated. What you cannot do is take the law anywhere that Cartesian coordinates do not exist.

And that is the consequence that decides the structure of this entire book. In general relativity spacetime is *curved*, and a curved space admits no global Cartesian coordinate system at all — not as a matter of difficulty but as a theorem (Chapter 3.4: the obstruction is the Riemann tensor, and it is nonzero exactly when curvature is present). There is no privileged frame in which the correction terms vanish everywhere, and so there is no frame in which $\mathbf{F} = m\mathbf{a}$ can be stated in its clean form. A law whose *shape* depends on using coordinates that do not exist cannot survive.

What is needed is a formulation that is form invariant from the outset — one that produces the correct equations in any coordinates, including ones nobody tabulated. Chapter 1.2 §7 proves that the Euler–Lagrange equations have exactly this property, for *arbitrary* smooth invertible coordinate changes, with a one-line reason: the action is a number attached to a path, and relabelling the points of a path cannot change a number. Chapter 3.2 then does gravity in that language, and (1.1.35)'s centrifugal and Coriolis terms reappear there as **Christoffel symbols** — the same artefacts, finally given a proper name and a geometric meaning.

4.3 · Failure three: fields

Newtonian mechanics is a theory of N particles. Its state is $6N$ numbers, its input is a list of pairwise forces, and its output is $3N$ coupled equations. Every part of that presupposes that N is finite.

An electromagnetic field is not like that. It has a value — six numbers, $\mathbf{E}$ and $\mathbf{B}$ — at *every point of space*, and there are uncountably many points. The configuration is not a list of numbers; it is a *function*, and the space of functions is infinite-dimensional (Chapter 0.4 built exactly that vector space, and Chapter 4.3 will take its infinite-dimensionality seriously).

So ask the Newtonian question of a field: what is the force on it? The question has no referent. There is no particle to apply $\mathbf{F} = m\mathbf{a}$ to, no mass to divide by, and no finite list to sum over. You can ask what force the field exerts on a charge — that is the Lorentz law — but the dynamics *of the field itself*, which is what Maxwell's equations describe, is simply not the kind of thing the second law can express. "Sum the forces on each particle and set the total equal to $m\mathbf{a}$ " has no meaning when there are no particles and no m.

You have already met the transition. Chapter 0.8 §7.6 took a chain of N masses coupled by springs, wrote the N coupled equations, and let $N \to \infty$ with the spacing going to zero; the N ordinary differential equations became one *partial* differential equation — the wave equation — and the N coordinates became a field $\phi(x, t)$. That limit is exactly where the particle description stops being useful and the field description takes over.

What is needed instead is a formulation whose input is **a single scalar function**, which does not care whether the thing it describes has three degrees of freedom or a continuum of them. That is a Lagrangian, and the extension to fields — replacing $L(q, \dot{q})$ by a Lagrangian *density* $\mathcal{L}(\phi, \partial_\mu \phi)$ integrated over space — is Chapter 5.2. Maxwell's equations then come from a single scalar (Chapter 2.6), as do Einstein's (Chapter 3.6) and the Standard Model's (Chapter 6.7). Not one of those can be written as $\mathbf{F} = m\mathbf{a}$.

4.4 · Failure four: quantum mechanics

The last failure is the most abrupt. In quantum mechanics, force is essentially not a concept.

⚑ **STATED NOW, PROVED IN PART IV**

In the standard formulation of quantum mechanics, every measurable quantity corresponds to a self-adjoint operator on a Hilbert space: position $\hat{x}$, momentum $\hat{p}$, angular momentum $\hat{\mathbf{L}}$, energy $\hat{H}$. **There is no force operator.** Nobody measures force on an electron, no textbook lists its eigenvalues, and it appears in no commutation relation.

What appears instead, everywhere, is the **Hamiltonian** $\hat{H}$ — the energy — which generates time evolution through $\mathrm{i}\hbar\,\partial_t\,|\psi\rangle = \hat{H}\,|\psi\rangle$ (Chapter 4.4), and whose classical ancestor is built from the Lagrangian by the Legendre transform of Chapter 1.3. The closest thing to a force is Ehrenfest's relation $\mathrm{d}\,\langle\hat{\mathbf{p}}\rangle\,/\mathrm{d}t = -\,\langle\nabla V\rangle$ (Chapter 4.6), and read carefully that is not a fundamental law but a *derived statement about expectation values*, in which the potential V is the primitive object and the force is what you get by differentiating it.

The same is true one level up. The path integral of Chapter 5.8 weights each history by $\mathrm{e}^{\mathrm{i}S/\hbar}$ with $S = \int L\,\mathrm{d}t$ — the action, again the Lagrangian. Quantum field theory is specified by writing down a Lagrangian density (Chapter 5.3), and the entire Standard Model fits on a T-shirt because it is one $\mathcal{L}$ (Chapter 6.7).

Take that as the practical argument, if the structural ones have not landed. **The objects that survive into the rest of physics are the Lagrangian and the Hamiltonian. The force does not.** Every hour spent on Chapters 1.2 and 1.3 is spent on machinery that reappears in Chapter 4.2, in Chapter 5.3, and in Chapter 6.7. Time spent perfecting free-body diagrams is not.

5 · The alternative, glimpsed: Fermat's principle

Everything so far has been demolition. Here is one small, complete, correct piece of physics that works in a completely different way — and once you have seen it, Chapter 1.2 is just the general theory of the same move.

It is about light, and it is older than Newton. Hero of Alexandria knew that a light ray reflecting off a mirror takes the *shortest* path; Fermat, in 1662, guessed the right generalisation:

FERMAT'S PRINCIPLE

Light travelling between two points takes the path for which the total travel time is stationary with respect to small deformations of the path.

Read what kind of statement that is, because it is unlike anything in §1–§4. It says nothing about what happens to the ray at any particular place. It does not describe a local push. It assigns a *single number* — the total travel time — to *each entire candidate path*, and then selects the path by a condition on that number. It is a statement about whole histories, not about instants.

5.1 · Snell's law, derived

Two media meeting at a flat horizontal interface. Light travels at speed $v_1 = c/n_1$ above and $v_2 = c/n_2$ below, where n is the refractive index. Place the source A at height a above the interface and the target B at depth b below it, with horizontal separation d.

Within each medium the speed is constant, so within each medium the fastest route between two points is a straight line — there is nothing to trade off. Therefore any candidate path is completely specified by *one number*: the horizontal position x at which it crosses the interface. Set the origin directly below A. Then the two legs have lengths $\sqrt{a^2 + x^2}$ and $\sqrt{b^2 + (d - x)^2}$ by Pythagoras, and the total time is

$$T(x) = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (d - x)^2}}{v_2}. \tag{1.1.37}$$

An infinite-dimensional problem — choose a path — has collapsed to an ordinary one-variable minimisation, and we can use Chapter 0.1. Differentiate, using the chain rule on each square root:

$$T'(x) = \frac{1}{v_1} \cdot \frac{x}{\sqrt{a^2 + x^2}} - \frac{1}{v_2} \cdot \frac{d - x}{\sqrt{b^2 + (d - x)^2}}. \tag{1.1.38}$$

Now recognise the two fractions. Let θ_1 be the angle the incident ray makes with the *normal* to the interface, and θ_2 the same for the refracted ray. In the right triangle with vertical leg a and horizontal leg x, the hypotenuse is the ray and the side opposite θ_1 is x, so

$$\sin\theta_1 = \frac{x}{\sqrt{a^2 + x^2}}, \qquad \sin\theta_2 = \frac{d - x}{\sqrt{b^2 + (d - x)^2}}. \tag{1.1.39}$$

These are not approximations; they are the definition of the sine applied to the two triangles. Substituting into (1.1.38) and imposing stationarity $T'(x) = 0$:

$$\frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}. \tag{1.1.40}$$

Finally put $v_i = c/n_i$, so $1/v_i = n_i/c$, and cancel the c:

$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2.} \tag{1.1.41}$$

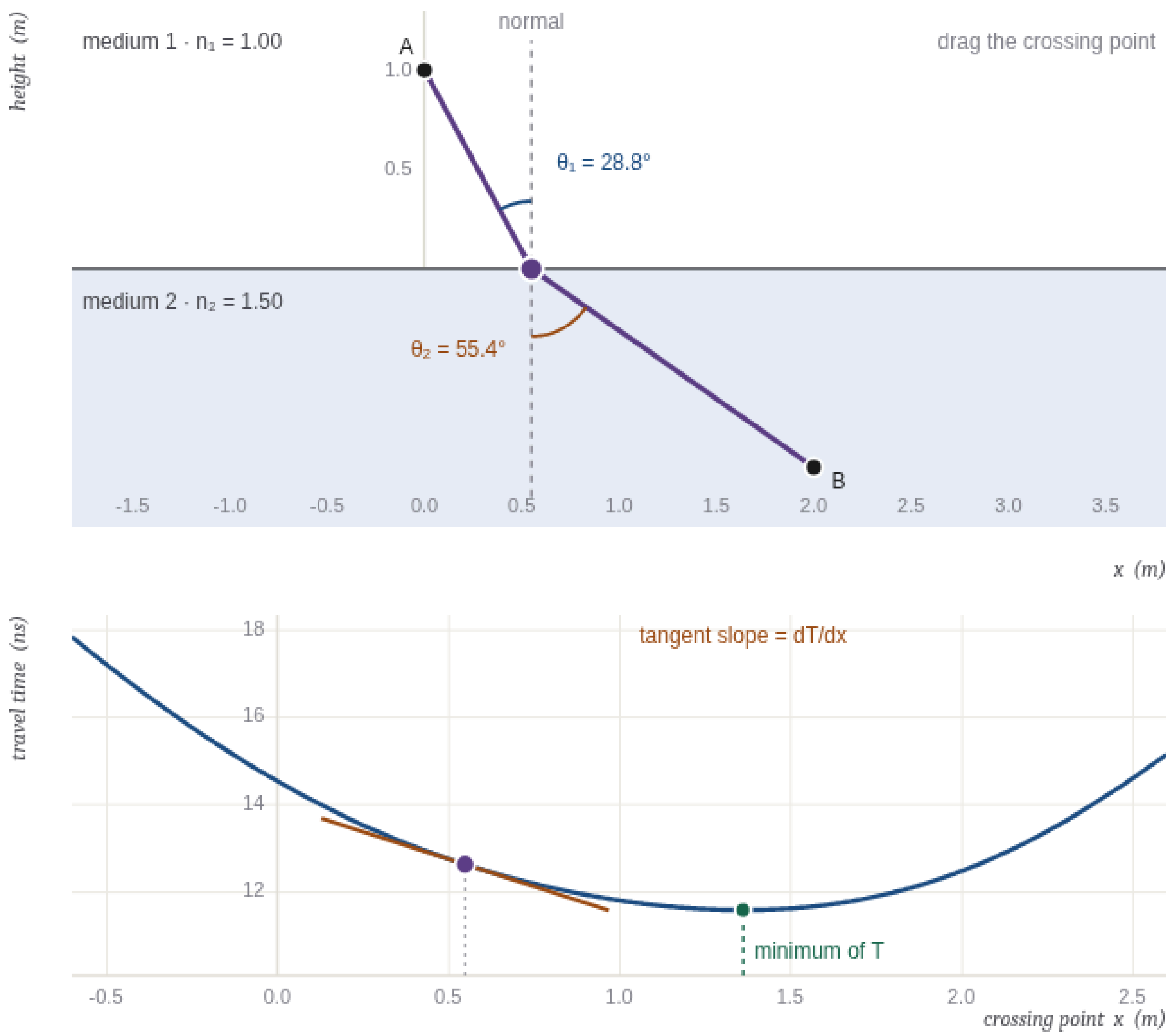
Snell's law, the empirical rule of refraction found by Ibn Sahl in 984 and rediscovered by Snellius in 1621 — derived, in half a page, from a statement about entire paths that mentions neither angles nor interfaces.

Two details before the figure. First, the stationary point is a genuine minimum here, and unique: differentiating (1.1.38) once more gives

$$T''(x) \;=\; \frac{a^2}{v_1\big(a^2+x^2\big)^{3/2}} \;+\; \frac{b^2}{v_2\big(b^2+(d-x)^2\big)^{3/2}} \;>\; 0 \tag{1.1.42}$$

everywhere, so T is strictly convex and has exactly one stationary point. Second, notice that (1.1.41) is a *local* statement — it constrains the two angles at one point, the crossing point — whereas Fermat's principle was a *global* statement about the whole path. Something has been converted.

5.2 · Watch it happen



crossing point *x* `x = 0.5500 m`

index n_1 `1.00` **index n_2** `1.50`

jump to the minimum of *T*

```
n₁ sin θ₁ =  0.48192          n₂ sin θ₂ =  1.23482

difference =  -0.75290             dT/dx =  -2.5114 ns/m

T = 12.619925 ns (minimum 11.571240 ns at x = 1.3627 m)
```

Snell's law is not assumed anywhere in this figure. Drag the crossing point along the interface (drag on either panel, or use the slider). The upper panel draws the two-leg ray with the normal dashed and the two angles marked; the lower panel plots the total travel time (1.1.37) as a function of the crossing point, with a short tangent line at your current position. Everything is computed numerically from (1.1.37) alone — the travel time by summing the two legs, the slope $\mathrm{d}T/\mathrm{d}x$ by a central difference, the marked minimum by a ternary search. Snell's law is never evaluated.

Now watch the two readouts $n_1 \sin\theta_1$ and $n_2 \sin\theta_2$ as you drag. They are generally different, and they *converge on each other* exactly as the tangent flattens. Press **jump to the minimum of *T*** and they agree to all five decimals, with $\mathrm{d}T/\mathrm{d}x = 0$. **Snell's law is not an extra law of optics; it is the stationarity condition, observed.** Push n_2 below n_1 and the ray bends the other way — away from the normal instead of toward it — because (1.1.41) now needs the larger sine on the far side. Note also how *flat* the time curve is near its minimum: a 5 cm error in the crossing point costs about 0.005 ns out of 11.6, which is Chapter 0.6's statement that at a stationary point the first-order change vanishes and the leading error is quadratic.

5.3 · The template

Step back from the optics and look at the shape of what happened, because that shape is the subject of Chapter 1.2.

THE MOVE

A global statement about a whole path — one number per path, made stationary — reproduced a local law holding at every point.

That is the entire template. Chapter 1.2 replaces "travel time" by a general integral $S = \int L\,\mathrm{d}t$, replaces "vary the crossing point" by "vary the whole function", and replaces "$T'(x) = 0$" by the Euler–Lagrange equation. The local law that drops out is the equation of motion.

Two features of the method deserve to be named, because they are precisely the two failures of §4.1 and §4.2.

It is coordinate-free. Nothing in Fermat's principle mentions x, y, or any axis. "The travel time along this path" is a property of the path and the media; you may compute it in whatever coordinates you like and you will get the same number, because it is a number. Contrast §4.2, where the *form of the law itself* changed when we relabelled the plane. A principle stated as "make this number stationary" cannot suffer that, and Chapter 1.2 §7 turns the observation into a theorem.

It needs only one scalar function. The entire input was $n(\mathbf{x})$ — one number at each point of space, from which the travel time of any path follows by integration. There were no vectors, no components, and no free-body diagrams. Contrast §4.3, where the force picture had nothing to say about a system with a continuum of degrees of freedom. A formulation whose input is a single scalar does not care how many degrees of freedom there are.

⚠ WHY THIS ISN'T OBVIOUS — "LEAST" IS THE WRONG WORD

Fermat's principle is usually quoted as the principle of *least* time, and (1.1.42) shows that for two flat media it really is a minimum. That is a coincidence of this example. The condition we actually imposed was $T'(x) = 0$, which by Chapter 0.6 signals a minimum, a maximum, or a saddle — and all three occur in optics.

The standard counterexample is a **concave mirror**. Put a source and a detector at the two foci of an ellipse and silver the ellipse: every path reflecting off the surface has *exactly the same* length, so T is constant and every path is stationary. Now replace the ellipse by a mirror that curves *more sharply* than that ellipse, still tangent at the reflection point. Every neighbouring path now reflects off a surface lying inside the ellipse, hence is shorter than the ellipse's constant value, so the actual ray — the one obeying the law of reflection — is a **local maximum** of the travel time. The physics is unchanged and the ray is exactly where it always was; only the word "least" was wrong.

So the correct statement is *stationary*, and the same correction will be needed for the "principle of least action" in Chapter 1.2 §5, where the space of paths is infinite-dimensional and saddle points are not the exception but the rule. Chapter 5.8 explains why stationarity is the right condition and minimisation is not: in the path integral, contributions from neighbouring paths cancel by interference wherever the action is *changing* at first order, and survive wherever it is not. Constructive interference cares that the phase is stationary. It does not care whether it is small.

6 · Worked examples

WORKED EXAMPLE 1 — FERMAT AND SNELL, WITH ACTUAL NUMBERS

Take $a = b = 1.000$ m, $d = 2.000$ m, air above ($n_1 = 1.00$) and glass below ($n_2 = 1.50$), with $c = 0.299792458$ m ns^{-1} so that times come out in nanoseconds. Find the path *without using Snell's law*, then check whether Snell's law happens to hold on it.

Step 1 — write the time. From (1.1.37),

$$T(x) = \frac{1.00\sqrt{1+x^2}}{0.299792458} + \frac{1.50\sqrt{1+(2-x)^2}}{0.299792458} \text{ ns.}$$

Step 2 — minimise numerically. Solving $T'(x) = 0$ by bisection on (1.1.38) to machine precision gives

$$x^\star = 1.362650205535 \text{ m}, \qquad T(x^\star) = 11.571240374 \text{ ns.}$$

The straight line from A to B crosses at $x = 1.000$ and takes 11.793271684 ns — slower by 0.222 ns. The light does *not* go straight, and it does not go by the shortest route either; it spends extra distance in the fast medium to shorten its stay in the slow one.

Step 3 — now check Snell. At $x^\star$, from (1.1.39),

$$\theta_1 = 53.726393^\circ, \quad n_1 \sin\theta_1 = 0.80620090813706,$$
$$\theta_2 = 32.511391^\circ, \quad n_2 \sin\theta_2 = 0.80620090813706.$$

Equal to fourteen digits; the difference is 2×10^{-16}, i.e. one unit in the last place of a double. Snell's law was not put in, and it came out.

Step 4 — confirm it is a minimum, and see how flat. (1.1.42) gives $T''(x^\star) = 3.6913$ ns m$^{-2} > 0$. Displacing the crossing point by $\pm\delta$:

δ (M)	$T(x^\star+\delta) - T(x^\star)$ (NS)	$\Delta T/\delta^2$
0.100	$+1.899 \times 10^{-2}$	1.8989
0.050	$+4.680 \times 10^{-3}$	1.8719
0.025	$+1.162 \times 10^{-3}$	1.8586

The excess is proportional to δ^2, and $\Delta T/\delta^2$ is converging on $\frac{1}{2}T''(x^\star) = 1.8457$ — Chapter 0.1's quadratic-error statement, measured. *The first order genuinely vanishes.* That is what "stationary" means, and it is the only property the argument used.

What to take from this. We never wrote down a law of refraction. We wrote down a number attached to each candidate path, differentiated it, and set the derivative to zero. The law of refraction was the output. Chapter 1.2

does the same thing when the "candidate path" is a whole function rather than a single number x, and the output is Newton's second law.

WORKED EXAMPLE 2 — ESCAPE VELOCITY, AND WHY ENERGY BEATS FORCE HERE

How fast must a projectile leave the surface of a body of mass M and radius R to never return?

The force route is impractical. You would integrate $m\ddot{r} = -GMm/r^2$, a nonlinear second-order ODE whose right-hand side depends on the unknown $r(t)$, and then take a limit as $t \to \infty$. Try it and you will be at it for a while.

The energy route is four lines. Newtonian gravity on a particle of mass m is

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}.$$

Is it conservative? Test (D): its curl vanishes (Problem 1 does the computation for exactly this form), and its domain $\mathbb{R}^3 \setminus \{\mathbf{0}\}$ is simply connected — in three dimensions you can always slide a loop off a removed point. So a potential exists, and since $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$,

$$V(r) = -\frac{GMm}{r} \qquad \Longrightarrow \qquad -\nabla V = -\frac{GMm}{r^2}\hat{\mathbf{r}} = \mathbf{F}.\ \checkmark$$

By (1.1.8), $E = \frac{1}{2}mv^2 - GMm/r$ is constant along the entire trajectory, however complicated. "Never returns" means r can grow without bound, and since $\frac{1}{2}mv^2 \geq 0$ always while $-GMm/r \to 0^-$ as $r \to \infty$, this is possible exactly when $E \geq 0$. Evaluating E at launch, $r = R$ and $v = v_e$:

$$\tfrac{1}{2}mv_e^2 - \frac{GMm}{R} = 0 \qquad \Longrightarrow \qquad \boxed{v_e = \sqrt{\frac{2GM}{R}}}$$

For the Earth ($G = 6.6743 \times 10^{-11}$, $M = 5.9722 \times 10^{24}$ kg, $R = 6.371 \times 10^6$ m) this is $11\,186$ m s^{-1} $= 11.19$ km s^{-1}; for the Moon, 2.375 km s^{-1}. Equivalently, since $g = GM/R^2$ at the surface, $v_e = \sqrt{2gR}$.

Three things the energy route gave you for free. The mass m cancelled, so escape velocity is the same for a pebble and a spacecraft. The *direction* of launch never entered, because E depends on $v^2 = \mathbf{v} \cdot \mathbf{v}$ and not on $\mathbf{v}$ — so (ignoring the ground and the atmosphere) sideways works as well as straight up, which is not remotely obvious from the force picture. And you never needed the trajectory at all: conservation replaced integration.

That last point is the general moral. A conserved quantity is a *first integral* — one integration of the equation of motion, already done, for free. Chapter 1.4 shows where they all come from, and by then "look for the conserved quantities first" will be the standard opening move.

7 · Your turn

PROBLEM 1 — IS THE INVERSE-SQUARE FORCE CONSERVATIVE?

Let $\mathbf{F} = -k\,\mathbf{r}/r^3$ with $r = |\mathbf{r}|$ and k constant — the shape of both gravity and the Coulomb force. (a) Compute $\nabla \times \mathbf{F}$ directly. (b) Find V with $\mathbf{F} = -\nabla V$. (c) State carefully why the domain matters, and why the conclusion here is *not* spoiled by the counterexample of Chapter 0.7 §2.4.

− **Solution**

(a) Write $F^x = -kx\,r^{-3}$ and use $\partial_x r = x/r$ (from $r = \sqrt{x^2+y^2+z^2}$). Then by the chain rule $\partial_x\big(r^{-3}\big) = -3r^{-4}\cdot(x/r) = -3x\,r^{-5}$, and

$$\partial_x F^y = \partial_x\big(-ky\,r^{-3}\big) = -ky\cdot\big(-3x\,r^{-5}\big) = \frac{3kxy}{r^5}.$$

By the symmetry of the expression in $x \leftrightarrow y$, $\partial_y F^x = 3kxy/r^5$ as well, so $(\nabla\times\mathbf{F})_z = \partial_x F^y - \partial_y F^x = 0$. The other two components follow by cycling $x \to y \to z \to x$, which the expression is invariant under. So $\nabla\times\mathbf{F} = \mathbf{0}$ on all of $\mathbb{R}^3 \setminus \{\mathbf{0}\}$. (Verified symbolically.)

(b) Guess a radial potential $V = V(r)$. Then $\nabla V = V'(r)\,\hat{\mathbf{r}} = V'(r)\,\mathbf{r}/r$, so we need $-V'(r)/r = -k/r^3$, i.e. $V'(r) = k/r^2$, giving

$$V(r) = -\frac{k}{r} + \text{const.}$$

Check: $-\nabla(-k/r) = k\nabla(1/r) = k\cdot\big(-\hat{\mathbf{r}}/r^2\big) = -k\mathbf{r}/r^3 = \mathbf{F}$. ✓ The additive constant is free, which is why "the potential at infinity is zero" is a choice and not a fact.

(c) Condition (D) holds, but (D)⇒(C) needs the domain to be simply connected, and $\mathbf{F}$ is undefined at the origin, so the domain is punctured. The saving fact is *dimension*: puncturing $\mathbb{R}^3$ leaves a simply connected region, because any loop can be slid sideways off the missing point and then contracted. Puncturing $\mathbb{R}^2$ does not — a loop encircling the hole is trapped, which is exactly why Chapter 0.7's vortex field $(-y,x)/(x^2+y^2)$ has zero curl and circulation 2π. Same puncture, different dimension, opposite conclusion. Here the theorem applies, the potential exists globally, and gravity is conservative everywhere it is defined.

Worth adding: $\nabla\cdot\mathbf{F} = 0$ away from the origin too (also verified symbolically). Both the divergence and the curl vanish on the whole domain, yet the field is emphatically not zero — all of its source sits in the single removed point, which is where Chapter 0.9's delta function goes, and $\nabla^2(1/r) = -4\pi\delta^3(\mathbf{r})$ is the statement that makes Gauss's law work.

PROBLEM 2 — THE TWO-BODY PROBLEM BECOMES A ONE-BODY PROBLEM

Two particles interact only with each other, by a force obeying the strong third law. (a) Show the centre of mass moves uniformly. (b) Show the relative coordinate obeys a one-body equation, and identify the mass appearing in it. (c) Show the total kinetic energy splits cleanly into the two pieces. (d) Compute the correction for hydrogen and say what it predicts.

– Solution

(a) With $\mathbf{F}_{12}$ the force on 1 from 2, $m_1\ddot{\mathbf{r}}_1 = \mathbf{F}_{12}$ and $m_2\ddot{\mathbf{r}}_2 = \mathbf{F}_{21} = -\mathbf{F}_{12}$. Adding, the right-hand sides cancel. Define $M = m_1 + m_2$ and $\mathbf{R} = (m_1\mathbf{r}_1 + m_2\mathbf{r}_2)/M$; then $M\ddot{\mathbf{R}} = \mathbf{0}$, so $\mathbf{R}(t) = \mathbf{R}_0 + \mathbf{V}t$. Three of the six degrees of freedom are now solved, exactly, forever. That is (1.1.10) in its most useful form.

(b) Let $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. Divide each equation by its mass and subtract:

$$\ddot{\mathbf{r}} = \frac{\mathbf{F}_{12}}{m_1} - \frac{\mathbf{F}_{21}}{m_2} = \mathbf{F}_{12}\left(\frac{1}{m_1} + \frac{1}{m_2}\right) \implies \mu\,\ddot{\mathbf{r}} = \mathbf{F}_{12},$$

where the **reduced mass** is defined by $1/\mu = 1/m_1 + 1/m_2$, i.e. $\mu = m_1 m_2/(m_1 + m_2)$. By the strong third law $\mathbf{F}_{12}$ points along $\mathbf{r}$ and (for the usual forces) depends only on r, so this is a single particle of mass μ in a central field. The two-body problem is solved.

(c) Inverting the definitions, $\mathbf{r}_1 = \mathbf{R} + (m_2/M)\mathbf{r}$ and $\mathbf{r}_2 = \mathbf{R} - (m_1/M)\mathbf{r}$. Then

$$\begin{aligned} T &= \tfrac{1}{2}m_1\left|\dot{\mathbf{R}} + \tfrac{m_2}{M}\dot{\mathbf{r}}\right|^2 + \tfrac{1}{2}m_2\left|\dot{\mathbf{R}} - \tfrac{m_1}{M}\dot{\mathbf{r}}\right|^2 \\ &= \tfrac{1}{2}M\dot{R}^2 + \dot{\mathbf{R}}\cdot\dot{\mathbf{r}}\,\frac{m_1 m_2 - m_2 m_1}{M} + \tfrac{1}{2}\dot{r}^2\,\frac{m_1 m_2^2 + m_2 m_1^2}{M^2} \\ &= \tfrac{1}{2}M\dot{R}^2 \;+\; \tfrac{1}{2}\mu\,\dot{r}^2. \end{aligned}$$

The cross terms cancel exactly — which is *why* $\mathbf{R}$ was defined with those weights — and the last step used $m_1 m_2(m_1 + m_2)/M^2 = \mu$. Centre-of-mass motion and relative motion do not talk to each other, in the equations or in the energy.

(d) For hydrogen, $\mu/m_e = m_p/(m_p + m_e) = 0.999455679$, a reduction of 5.44×10^{-4}. Every energy level scales with μ, so every spectral line shifts by that fraction. For deuterium the nucleus is about twice as heavy, giving $\mu/m_e = 0.999727630$, and the fractional difference between the two is 2.721×10^{-4}. Applied to the H-α line at 656.28 nm this predicts a deuterium line shifted by 0.179 nm toward the blue — which is how Urey identified deuterium in 1932, and it is a purely classical centre-of-mass effect sitting inside a quantum calculation.

PROBLEM 3 — POLAR ACCELERATION, AND WHICH TERMS ARE REAL

(a) Derive (1.1.35) by a route that never mentions unit vectors: write $x = r\cos\theta, y = r\sin\theta$, differentiate twice, and project. (b) A particle moves on a circle of fixed radius R at constant angular rate ω. Compute F_r and F_θ. (c) Which of the four terms in (1.1.35) would survive a change to a different coordinate system, and which are artefacts?

– Solution

(a) Differentiate $x = r\cos\theta$ twice with the product and chain rules:

$$\dot{x} = \dot{r}\cos\theta - r\dot{\theta}\sin\theta,$$
$$\ddot{x} = \ddot{r}\cos\theta - 2\dot{r}\dot{\theta}\sin\theta - r\ddot{\theta}\sin\theta - r\dot{\theta}^2\cos\theta,$$

and similarly $\ddot{y} = \ddot{r}\sin\theta + 2\dot{r}\dot{\theta}\cos\theta + r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta$. The radial component of the acceleration is $a_r = \ddot{x}\cos\theta + \ddot{y}\sin\theta$; every term either picks up $\cos^2 + \sin^2 = 1$ or the cancelling combination $-\cos\sin + \sin\cos = 0$:

$$a_r = \ddot{r} - r\dot{\theta}^2.$$

The tangential component is $a_\theta = -\ddot{x}\sin\theta + \ddot{y}\cos\theta$, and the same bookkeeping gives $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$. (Both confirmed with a computer algebra system.) Multiply by m and you have (1.1.35). Note this route is *longer*, not shorter — the unit-vector derivatives of (1.1.32) were doing real work.

(b) $r = R$ constant, so $\dot{r} = \ddot{r} = 0$; $\dot{\theta} = \omega$ constant, so $\ddot{\theta} = 0$. Then $F_r = -mR\omega^2$ and $F_\theta = 0$. The force is purely radial and points *inward* — the centripetal force, whose magnitude $mR\omega^2 = mv^2/R$ falls out of the general formula rather than being quoted. There is no outward "centrifugal force" acting on the particle; the $-r\dot{\theta}^2$ sits on the acceleration side of the equation.

(c) None of the four terms is coordinate-independent, and that is the point. Under a change of coordinates the whole left-hand side transforms as a unit; individual terms have no invariant meaning. But there is a sharp version of the question. Write the equations in Cartesian coordinates and the extra terms are *absent*; write them in polar and they are *present*; the physics is identical. Any term you can remove by relabelling the plane is not describing a physical interaction, so **$-mr\dot{\theta}^2$ and $2m\dot{r}\dot{\theta}$ are artefacts of the labelling.** (1.1.36) makes it concrete: a free particle has $\ddot{r} \neq 0$.

The right way to say this, available from Chapter 3.3 onward, is that $\ddot{q}^i + \Gamma^i{}_{jk}\dot{q}^j\dot{q}^k$ is a coordinate-independent object while $\ddot{q}^i$ alone is not, and the Γ's are exactly the centrifugal and Coriolis terms. Gravity, remarkably, turns out to belong to the same category: it too can be removed at a point by choosing the right coordinates (a freely falling frame), which is the equivalence principle of Chapter 3.1. What distinguishes gravity from centrifugal force is that *no single* coordinate change removes it everywhere at once, and the obstruction is curvature.

PROBLEM 4 — THE THIRD LAW FAILS, QUANTITATIVELY

Two protons ($q = 1.602 \times 10^{-19}$ C) are separated by $d = 1.00$ nm along $\hat{\mathbf{x}}$. Proton 1, at the origin, moves with $\mathbf{v}_1 = v\hat{\mathbf{x}}$; proton 2 moves with $\mathbf{v}_2 = v\hat{\mathbf{y}}$, where $v = 3.00 \times 10^6$ m s^{-1}. (a) Compute the magnetic force on each. (b) Compute $\mathrm{d}\mathbf{P}_{\text{mech}}/\mathrm{d}t$ for the pair and compare to the Coulomb force. (c) Where is the missing momentum, and what would you have to compute to find it?

− Solution

(a) The field of 1 at 2 involves $\mathbf{v}_1 \times \hat{\mathbf{r}} = v\hat{\mathbf{x}} \times \hat{\mathbf{x}} = \mathbf{0}$, so $\mathbf{B}_1(\mathbf{r}_2) = \mathbf{0}$ and **the magnetic force on proton 2 is exactly zero.** For the other direction, $\hat{\mathbf{r}}$ from 2 to 1 is $-\hat{\mathbf{x}}$, so by (1.1.14)

$$\mathbf{B}_2(\mathbf{r}_1) = \frac{\mu_0 q v}{4\pi d^2}\,\hat{\mathbf{z}}, \qquad \mathbf{F}_{12}^{\text{mag}} = qv\hat{\mathbf{x}} \times \mathbf{B}_2 = -\frac{\mu_0 q^2 v^2}{4\pi d^2}\,\hat{\mathbf{y}}.$$

Numerically, with $\mu_0/4\pi = 10^{-7}$ exactly, $q^2 = 2.566 \times 10^{-38}$, $v^2 = 9.00 \times 10^{12}$ and $d^2 = 10^{-18}$,

$$|\mathbf{F}_{12}^{\text{mag}}| = \frac{10^{-7} \times 2.566 \times 10^{-38} \times 9.00 \times 10^{12}}{10^{-18}} = 2.31 \times 10^{-14}\,\text{N}.$$

(b) The Coulomb forces are equal and opposite and cancel in the sum, so the net internal force on the pair is the magnetic residue alone:

$$\frac{\mathrm{d}\mathbf{P}_{\text{mech}}}{\mathrm{d}t} = \mathbf{F}_{12} + \mathbf{F}_{21} = -2.31 \times 10^{-14}\,\hat{\mathbf{y}}\,\text{N}.$$

The Coulomb force is $q^2/4\pi\varepsilon_0 d^2 = 8.988 \times 10^9 \times 2.566 \times 10^{-38}/10^{-18} = 2.307 \times 10^{-10}$ N, so the ratio is 1.00×10^{-4}. Check it against the closed form (1.1.18): both forces carry the same q^2/d^2, so the ratio must be $\mu_0\varepsilon_0 v^2 = v^2/c^2 = \left(3.00 \times 10^6/2.998 \times 10^8\right)^2 = 1.00 \times 10^{-4}$. ✓ The d-dependence cancels, so the violation is the same fractional size at any separation.

The conclusion stands regardless of the numbers: an isolated pair of particles has $\mathrm{d}\mathbf{P}/\mathrm{d}t \neq \mathbf{0}$. The effect is suppressed by v^2/c^2 — which is why nobody noticed for two centuries, and exactly why it is a relativistic effect in disguise.

(c) In the electromagnetic field. To find it you would compute $\mathbf{P}_{\text{field}} = \varepsilon_0 \int \mathbf{E} \times \mathbf{B}\,\mathrm{d}^3x$ over all space, using the total $\mathbf{E}$ and $\mathbf{B}$ of both charges, and verify that $\mathrm{d}(\mathbf{P}_{\text{mech}} + \mathbf{P}_{\text{field}})/\mathrm{d}t = \mathbf{0}$. Chapter 2.6 derives the integrand from Maxwell's equations and proves the conservation law in general; the mechanism is the **Maxwell stress tensor**, which is the momentum flux through a surface, and the proof is the field-theoretic version of (1.1.10).

A further honesty note: the Biot–Savart expression we used is the leading term in v/c. A full treatment uses the Liénard–Wiechert fields, which are *retarded* — the field at charge 2 now depends on where charge 1 was d/c seconds ago. Retardation makes the failure of the third law inevitable rather than accidental: if the interaction is not instantaneous, "equal and opposite *at the same instant*" is not even a well-posed statement, since different observers disagree about which instants are simultaneous (Chapter 2.3).

THE BRICK YOU JUST LAID

You have Newton's three laws stated with their fine print: the first as a *definition* of the frames in which the second holds, the second as a second-order ODE whose Picard–Lindelöf uniqueness *is* determinism, the third in two strengths. You know that the second law is a framework and not a theory, because it does not say what $\mathbf{F}$ is.

You have **energy** — the work–energy theorem from one dot product, kinetic energy appearing unbidden, and $E = T + V$ conserved exactly when the force is conservative in Chapter 0.7's sense. That is the machinery Chapter 1.2 is built from ($L = T - V$), Chapter 1.3 rebuilds as the Hamiltonian, and Chapter 1.4 re-derives from time-translation symmetry.

You have **momentum and angular momentum** from the weak and strong third laws, and a crack: the third law is *false* for magnetic forces between moving charges, momentum is not conserved among the particles alone, and the only available repair is that **the field carries momentum** — which is the first evidence in this book that fields are objects rather than bookkeeping. That debt is paid in Chapter 2.6.

And you have **four demonstrated failures** of the force formulation, each with a chapter attached. **(a)** Constraints: a page of algebra and an unwanted tension for the pendulum — answered by Chapter 1.2 §6, in three lines. **(b)** Coordinates: $\mathbf{F} = m\mathbf{a}$ is not form invariant, and general relativity has no global Cartesian frame — answered by Chapter 1.2 §7 and spent in Chapter 3.2. **(c)** Fields: uncountably many degrees of freedom, no particles to sum forces over — answered by Chapter 5.2. **(d)** Quantum: there is no force operator, but there is a Hamiltonian — answered by Chapters 4.2 and 5.3.

Finally you have **Fermat's template**: one number per path, made stationary, yielding a local law at every point — coordinate-free, needing only a single scalar function, which are precisely the two things (a) and (b) lacked. Chapter 1.2 generalises it, and every remaining theory in this book is written in the language it produces.